Reflect with us - Week 5

This assignment will not be graded and is only for practice.

1 point

Consider a map $T: R^2 \to R^2$ defined as $T(v) = Av$, where $v = \begin{bmatrix} x \\ y \end{bmatrix}$, $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $det(A) \neq 0$. Which of the following options are correct?

☐ T is a linear transformation.

☐ T is both one-one and onto.

☐ T is neither one-one nor onto.

☐ T is one-one but not onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is a linear transformation.

T is both one-one and onto.

- Step 1:

  1 point

  What is $T(v_1 + v_2)$?

  ○ $A(v_1 + v_2)$.

  ○ $Av_1 v_2$.

  No, the answer is incorrect.

  Score: 0

  Accepted Answers:

  $A(v_1 + v_2)$.

  Recall: $A(v_1 + v_2) = Av_1 + Av_2$

- Step 2:

  1 point

  Is $T(v_1 + v_2) = T(v_1) + T(v_2)$?

  ○ Yes.

  ○ No.

  No, the answer is incorrect.

  Score: 0

  Accepted Answers:

  Yes.

  Check: $T(cv) = cT(v)$.

- Hence T is linear transformation. So Option 1 is true.
- Another method: Write the definition of T explicitly as,

$$T(x, y) = (ax + by, cx + dy)$$

- Step 1:Step 1: What is $T((x_1, y_1) + (x_2 + y_2))$?

  $$\begin{aligned} T(x_1 + x_2, y_1 + y_2) &= (a(x_1 + x_2) + b(y_1 + y_2), c(x_1 + x_2) + d(y_1 + y_2)) \\ &= (ax_1 + ax_2 + by_1 + by_2, cx_1 + cx_2 + dy_1 + dy_2) \\ &= (ax_1 + by_1, cx_1 + dy_1) + (ax_2 + by_2, cx_2 + dy_2) \\ &= T(x_1, y_1) + T(x_2, y_2) \end{aligned}$$

- Step 2:Step 2:

  1 point

  Is $T(c(x, y)) = cT(x, y)$?

  ○ Yes.
  ○ No.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  Yes.

- Hence T is linear transformation. So Option 1 is true.

  Now in the given problem $T(v) = 0$, implies $Av = 0$. This is the matrix representation of the system of linear equations:

  $$\begin{aligned} ax + by &= 0 \\ cx + dy &= 0 \end{aligned}$$

- Step 3:Step 3:

  1 point

  If the system of linear equations $Av = 0$ has a unique solution, then what is the solution?

  ○ $v = 0$

  ○ v can be any vector in R^2.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  $v = 0$

- Step 4:Step 4:

  1 point

  What is the condition on A, for which the system of linear equations $Av = 0$ has a unique solution? [Recall from Weeks 1 and 2]

  ○ $det(A) = 0$.

  ○ $det(A) \neq 0$.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  $det(A) \neq 0$.

  Concluding, we can say that, if $det(A)$ is non-zero, then $T(v) = 0$ implies $v = 0$. Hence T is one to one.

- Let $w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \in R^2$ w = [w1w2] ∈ R2. We are going to find whether there exists a vector

  $v = \begin{bmatrix} x \\ y \end{bmatrix} \in R^2$ v = [xy] ∈ R2, such that $T(v) = Av = w$T(v) = Av = w.

  $Av = w$Av = w is the matrix representation of the system of linear equation:

  $$\begin{aligned} ax + by &= w_1 \\ cx + dy &= w_2 \end{aligned}$$ ax + bycx + dy = w1 = w2

- Step 5:Step 5:

  1 point

  If $det(A) \neq 0$det(A) = 0, then what can we say about the solution of the above system of linear equations?

  ○ No solution.
  ○ Unique solution.

  No, the answer is incorrect.
  Score: 0
  Accepted Answers:
  Unique solution.
  Concluding from this, we can say that, if $det(A) \neq 0$det(A) = 0, then TT is onto.

  Hence, Options 1 and 2 are the correct options.Hence, Options 1 and 2 are the correct options.

  Check Answers

  Your score is: 0/7